Cuisine Classification Using Machine Learning

1. Objective

The objective of this project is to build a machine learning model that classifies restaurants based on their cuisine type using restaurant-related features such as location, cost, delivery options, and customer engagement metrics.

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2. Dataset Information

- Total Records: 9551
- Missing Values: 9 records in the Cuisines column
- Target Variable: Primary Cuisine
- Features Used:
    - Country Code
    - City
    - Average Cost for Two
    - Currency
    - Has Table Booking
    - Has Online Delivery
    - Is Delivering Now
    - Switch to Order Menu
    - Price Range
    - Votes

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3. Data Preprocessing

- Removed missing values from the Cuisines column.
- Extracted Primary Cuisine from cuisine combinations.
- Encoded categorical variables using Label Encoding.
- Split the dataset into training and testing sets (80:20).

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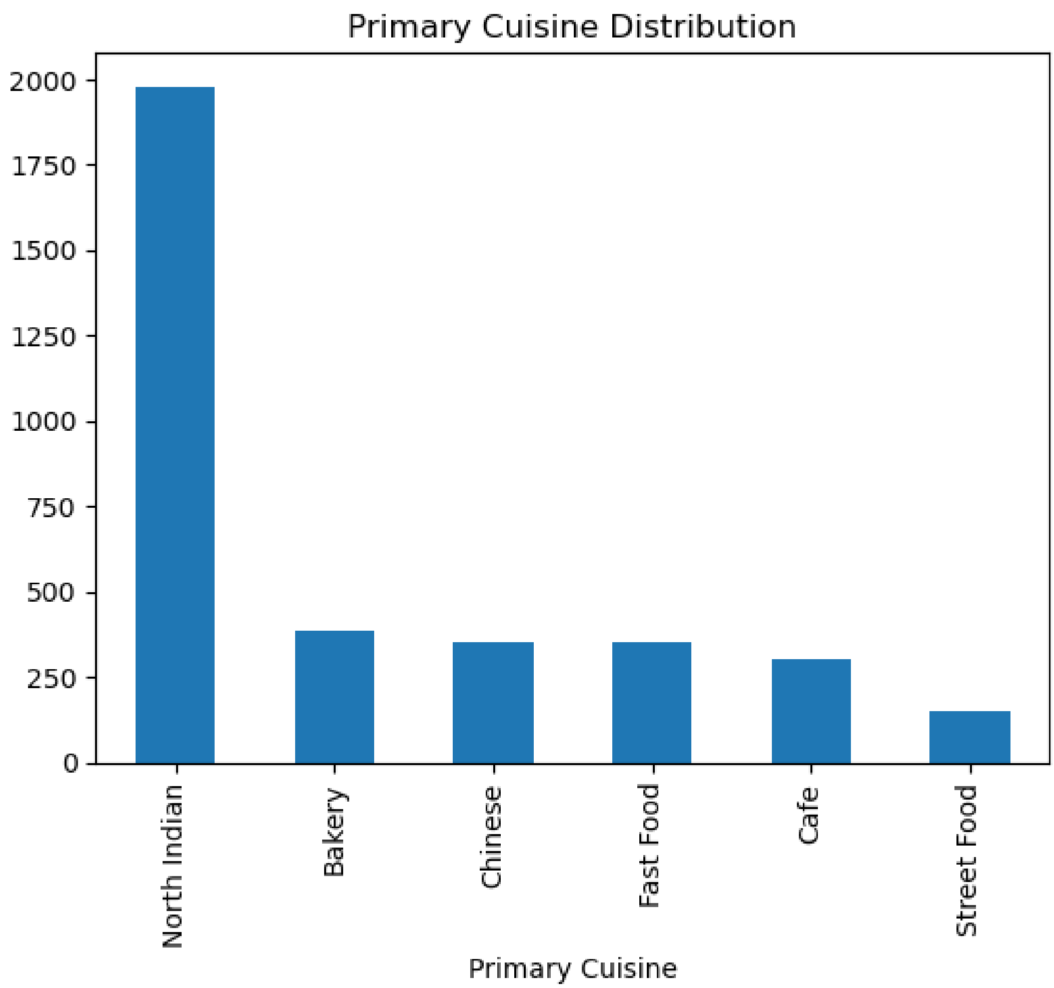
4. Exploratory Data Analysis

Cuisine Distribution



Observation:

- North Indian cuisine dominates the dataset.
- Significant class imbalance exists among cuisine categories.

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5. Model Training

Logistic Regression

A linear classification model used as a baseline classifier.

Random Forest Classifier

An ensemble learning model that combines multiple decision trees for classification.

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6. Model Evaluation

Final Results

Model	Accuracy
Logistic Regression	57.02%
Random Forest	52.34%

Observation:

- Logistic Regression achieved the highest accuracy.
- Both models performed significantly better after feature engineering.

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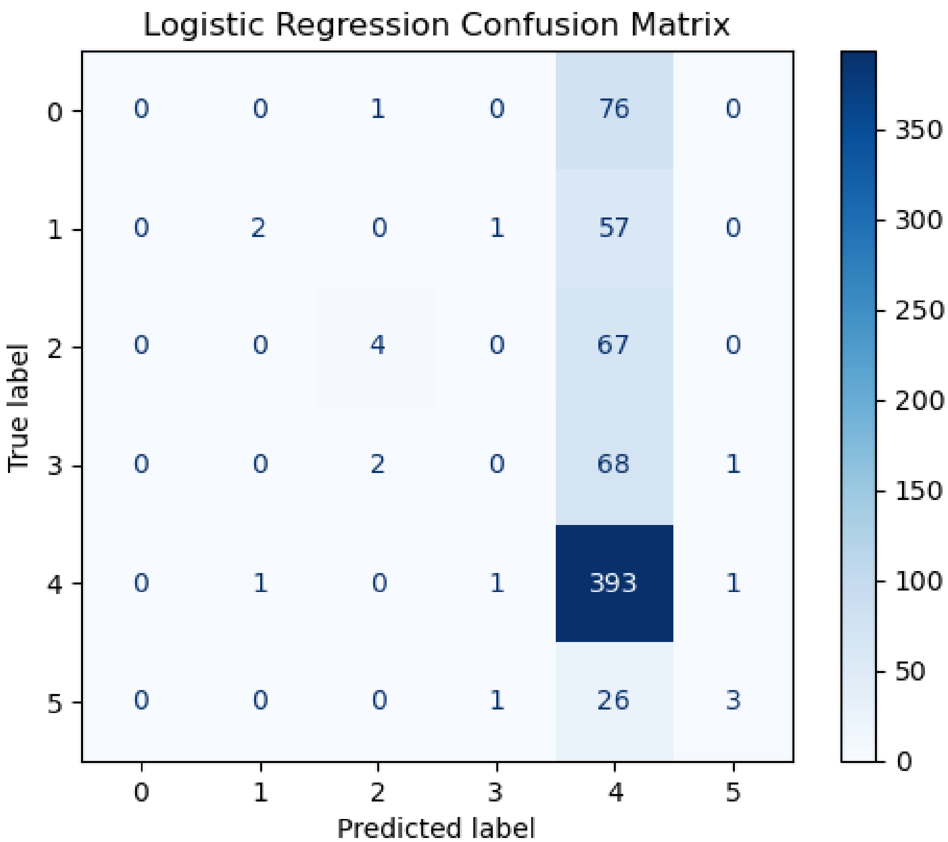
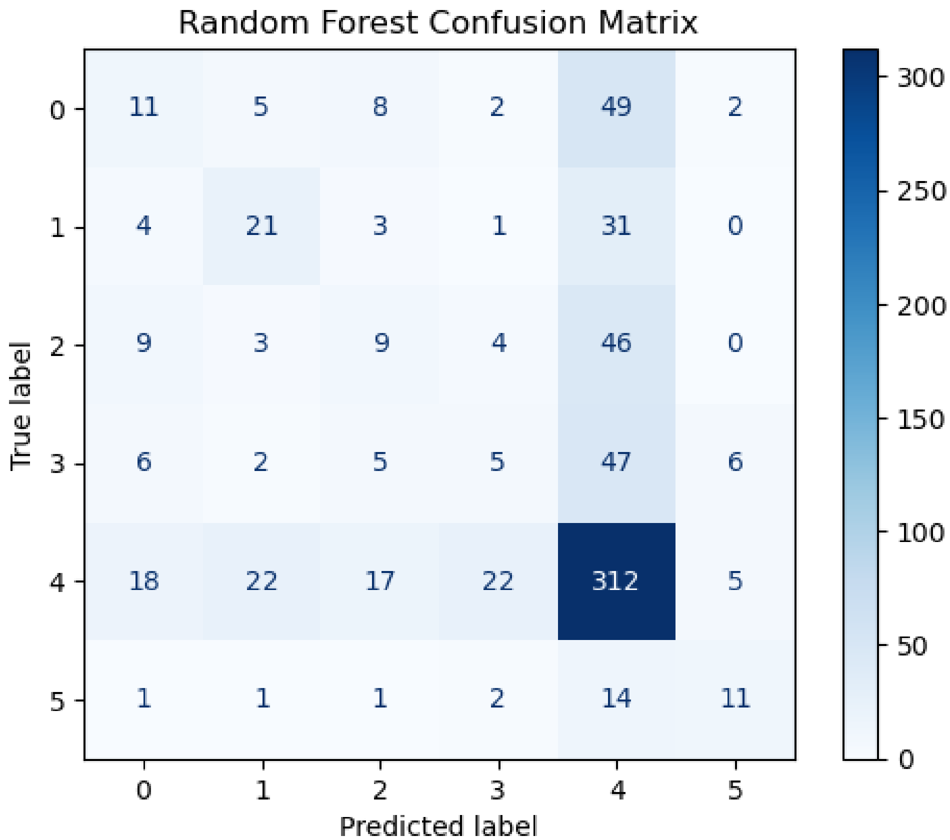
7. Confusion Matrix Analysis

Logistic Regression Confusion Matrix



Random Forest Confusion Matrix



Observation:

- Most predictions were concentrated around major cuisine categories.
- Misclassifications occurred mainly between similar cuisine groups.

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8. Model Improvement

Initially, cuisine combinations were used as target classes.

Initial Results

Model	Accuracy
Logistic Regression	31.21%
Random Forest	28.79%

After extracting the Primary Cuisine feature:

Improved Results

Model	Accuracy
Logistic Regression	57.02%
Random Forest	52.34%

Model	Accuracy	
Classification Model	Initial Accuracy (%)	Improved Accuracy (%)
Logistic Regression	31.21	57.02
Random Forest	28.79	52.34

Observation:

- Accuracy increased significantly after reducing overlapping cuisine classes.
- Feature engineering improved model performance by approximately 25 percentage points.

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9. Challenges and Biases

- The dataset contains class imbalance.
- Some cuisine categories have significantly fewer samples.
- Restaurant metadata alone may not perfectly determine cuisine type.
- Similar cuisines often share common restaurant characteristics, causing classification challenges.

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10. Conclusion

The Cuisine Classification System successfully classified restaurants into cuisine categories using machine learning techniques. Logistic Regression achieved the best performance with an accuracy of **57.02%**, outperforming Random Forest. Creating the Primary Cuisine feature significantly improved model accuracy and overall model effectiveness.